Airbnb and Crime: A Spatial Analysis of Socioeconomic Effects in Chicago

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1 Introduction

The relationship between short-term rental platforms like Airbnb and crime rates has drawn significant attention in recent years. Studies have shown that the prevalence of Airbnb listings can contribute to increased crime rates in urban neighborhoods. For example, research conducted in London has demonstrated that neighborhoods with more Airbnb activity experience higher rates of property crime and antisocial behavior [1].

While the association between Airbnb presence and crime has been documented, the relationship between Airbnb rental fees and crime rates remains less clear. Some hypotheses suggest that higher rental fees may attract more affluent tourists, potentially increasing the possibility of more theft or property crimes. However, empirical evidence to support this claim is limited, and the relationship may vary depending on neighborhood characteristics and socioeconomic conditions.

In this project, we aim to analyze the impact of Airbnb-related variables and socioeconomic indicators on crime patterns in Chicago. Specifically, we focus on the following explanatory variables: the number of Airbnb spots (`num_spots`), the price per person (`price_pp`), the poverty rate (`poverty`), and income per capita (`income_pc`). The response variables considered are the number of thefts (`num_theft`), the number of crimes (`num_crimes`), and the ratio of thefts to crimes (`rate`). To account for spatial dependencies, we fit Spatial Autoregressive (SAR) and Conditional Autoregressive (CAR) models. In designing these models, to manage the complexity of potential model candidates, we assumed that the effects of socioeconomic indicators and Airbnb-related factors on crime rates are independent. Additionally, when considering two variables within either the socioeconomic indicators or the Airbnb-related factors, we included their interaction term in the model design.

Our analysis aims to provide insights into how Airbnb presence and socioeconomic factors influence crime patterns, contributing to the broader understanding of urban crime dynamics and short-term rental activity.

2 Methods

2.1 Methods

2.1.1 Moran's I test

Spatial autocorrelation measures the degree to which objects that are close together in space exhibit similar values. Moran's I statistic is a widely used measure to detect spatial autocorrelation. The null hypothesis of the Moran's I test is that the observed spatial pattern is random, implying no spatial autocorrelation.

Formula for Moran's I:

$$I = \frac{m \sum_{i=1}^{m} \sum_{j=1}^{m} w_{ij}(y_i - \bar{y})(y_j - \bar{y})}{\left(\sum_i \sum_j w_{ij}\right) \sum_{i=1}^{m} (y_i - \bar{y})^2}$$

where m is the number of regions, y_i and y_j are the values at the locations i and j, $\bar{y}$ is the mean of y_i, w_{ij} are spatial proximity.

In this project, we use `moran.test` and `moran.mc`. We used the `moran.test` function to compute Moran's I and assess the significance of spatial autocorrelation. This function performs a hypothesis

test for spatial autocorrelation based on a weights matrix. To further validate the results, we performed a Monte Carlo permutation test using the `moran.mc` function. This test provides a simulated distribution of Moran's I by randomly permuting the data, which helps to determine the robustness of the observed spatial autocorrelation.

2.1.2 Simultaneous Auto-regressive (SAR) Model

Given the presence of spatial autocorrelation, we used a Spatial Autoregressive (SAR) model to account for spatial dependencies in our regression analysis. The SAR model incorporates spatial lags of the dependent variable, making it suitable for analyzing data where the outcome in one region is influenced by neighboring regions.

Formla for SAR:

$$\boldsymbol{Y} = \boldsymbol{X\beta} + \rho \boldsymbol{W}(\boldsymbol{Y} - \boldsymbol{X\beta}) + \boldsymbol{\epsilon}, \boldsymbol{\epsilon} \sim N(\boldsymbol{0}, \sigma^2 \boldsymbol{I})$$

where $\boldsymbol{W}$ contains the weights and $\boldsymbol{Y}, \boldsymbol{\epsilon}$ are n-by-1 vectors and $\boldsymbol{I}$ is the identity matrix, $\boldsymbol{X}$ is n-by-m matrix containing explanatory variables and $\boldsymbol{\beta}$ is an m-by-1 vector of the regression coefficients.

In this project, the `spautolm` function from the `spatialreg` package was used to fit the SAR model. The argument `family="SAR"` specifies that the SAR model is applied. The model was estimated using Maximum Likelihood Estimation (MLE), which ensures efficient and unbiased parameter estimation.

2.1.3 Conditional Autoregressive (CAR) Model

The Conditional Autoregressive (CAR) model defines spatial relationships through a set of conditional distributions. Specifically, the value at a given region depends on the values of its neighboring regions. This dependency is captured through the conditional distribution of the observation at one location, given the values at other locations. The CAR model can be expressed as:

$$E(x_i \mid x_{-i}) = \mu + \sum_{j \neq i} \beta_{ij}(x_j - \mu_j),$$

$$Q_{i|-i} = \frac{1}{\sigma_i^2} > 0,$$

where β_{ij} is the spatial dependence coefficient between regions i and j, $\beta_{ii} = 0$, $Q_{ii} = 1/\sigma_i^2$, and $-Q_{ij}/Q_{ii} = \beta_{ij}$. Here, σ^2 represents the conditional variance.

The ICAR model is a variant of the CAR model where the spatial dependence coefficient β_{ij} is proportional to the spatial weights W_{ij}. When we assume $\beta_{ij} = W_{ij}/W_{i+}$, where $W_{i+} = \sum_{j \neq i} W_{ij}$ and $1/\sigma_i^2 = \tau^2 W_{i+}$, the model becomes:

$$E(u_i \mid u_{-i}) = \sum_{j \neq i} \frac{W_{ij}}{W_{i+}} u_j,$$

$$Q_{i|-i} = \tau^2 W_{i+}, \quad \tau^2 > 0.$$

The precision matrix $\mathbf{Q}$ of the ICAR model is expressed as:

$$\mathbf{Q} = \tau^2(\mathbf{M} - \mathbf{W}),$$

where $\mathbf{M}$ is a diagonal matrix with entries $M_{ii} = W_{i+}$. Because $\mathbf{Q1} = \tau^2(\mathbf{M} - \mathbf{W})\mathbf{1} = 0$, the precision matrix $\mathbf{Q}$ is not of full rank. This property makes the joint distribution improper, a characteristic that defines the intrinsic CAR (ICAR) model.

In our project, we applied the ICAR model to account for spatial autocorrelation in crime data, ensuring that neighboring regions' interactions were properly modeled. We used the `spautolm` function from the `spatialreg` package to fit the CAR model. The argument `family='CAR'` specifies the use of an intrinsic Conditional Autoregressive (ICAR) model.

2.2 Data

The dataset used in this project is sourced from the **GeoDa Center for Geospatial Analysis and Computation** and contains information about Airbnb listings and crime statistics in Chicago for the year 2015. This dataset enables the spatial analysis of the relationship between short-term rental activity and crime patterns. Below is a brief description of the key variables used in the project:

- `num_theft` This variable represents the number of theft incidents recorded in a specific community area of Chicago. It is a count of reported theft crimes, including larceny and related property crimes.
- `num_crime` This variable indicates the total number of crimes (including various types such as theft, assault, burglary, etc.) reported in a community area. It provides an overall measure of criminal activity within the area.
- `num_spots` This variable captures the number of Airbnb listings or "spots" available in each community area.
- `price_pp` This variable represents the average price per person for an Airbnb rental in each community area.
- `poverty` The percentage of the population in a community area living below the poverty line.
- `income_pc` This variable measures the average income per capita for residents in each community area.

This dataset allows for the exploration of how Airbnb presence (number of listings and pricing) correlates with crime rates, while considering socioeconomic factors such as poverty and income levels. The selected variables provide a comprehensive view of the spatial relationships between short-term rentals and crime patterns in Chicago's neighborhoods.

Moran's I test To assess the presence of spatial autocorrelation in the response variables (`num_theft`, `num_crimes`, and `rate`), we conducted Moran's I tests. The Moran's I statistic evaluates whether spatial patterns in the data are clustered, dispersed, or random. Significant positive Moran's I values indicate clustering, suggesting spatial dependence.

We applied the `moran.test` function to each response variable, using a spatial weights matrix to define neighborhood relationships. Additionally, we performed a Monte Carlo simulation with 999 permutations using the `moran.mc` function to validate the results. The test results for Moran's I and the Monte Carlo simulation are presented in Tables 1 and 2, respectively.

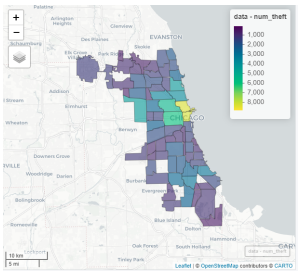
Table 1: Moran's I Test Results

Model	Std. Deviate	p-value	Moran's I	Expectation	Variance
Theft Model	6.9265	2.158×10^{-12}	0.5009	-0.0147	0.0055
Crime Model	4.9571	3.577×10^{-7}	0.3636	-0.0147	0.0058
Rate Model	7.4765	3.815×10^{-14}	0.5683	-0.0147	0.0061

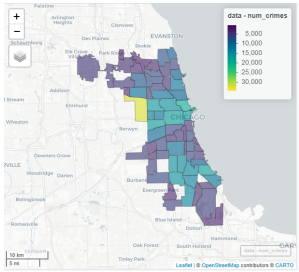
Table 2: Monte Carlo Simulation of Moran's I Results

Model	Statistic	Observed Rank	p-value
Theft Model	0.5009	1000	0.001
Crime Model	0.3636	1000	0.001
Rate Model	0.5683	1000	0.001

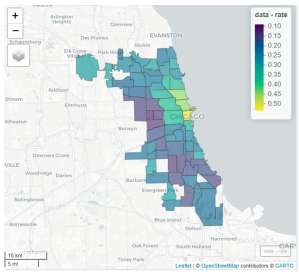
Visualization with `tmap` To better understand the spatial distribution of the response and explanatory variables, we used the `tmap` package for visualization. Figures 1 and 2 present choropleth maps showing the spatial patterns for thefts, crimes, and the theft-to-crime ratio, as well as Airbnb-related and socioeconomic indicators.



(a) Number of theft

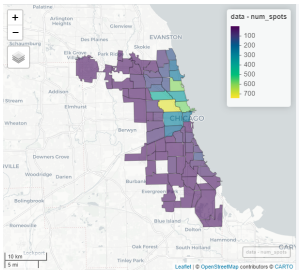


(b) Number of crime

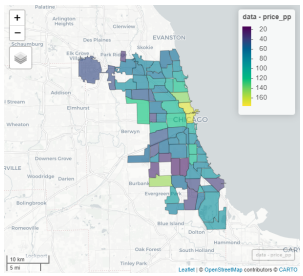


(c) Rate

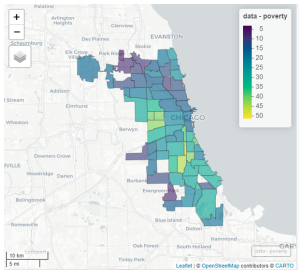
Figure 1: Mapview



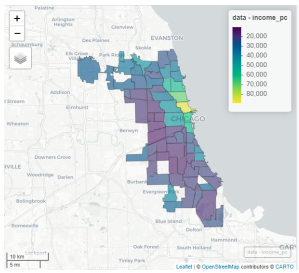
(a) Number of Airbnb Spots



(b) Price per Person



(c) Poverty Rate



(d) Income per Capita

Figure 2: Mapview

These visualizations reveal clear spatial patterns, suggesting clustering of crimes and socioeconomic indicators across Chicago neighborhoods.

3 Results

3.1 Model Fitting

To investigate the relationship between crimes and Airbnb-related factors in Chicago, we fitted several Spatial Autoregressive (SAR) and Conditional Autoregressive (CAR) models. Each model included different combinations of explanatory variables such as the number of Airbnb spots (`num_spots`), price per person (`price_pp`), poverty rate (`poverty`), and income per capita (`income_pc`). These models were fitted for three response variables: the number of thefts (`num_theft`), the number of crimes (`num_crimes`), and the ratio of thefts to crimes (`rate`).

To simplify the model selection process, we assumed that the effects of socioeconomic indicators and Airbnb-related factors on crime rates are independent. Additionally, within each category (socioeconomic or Airbnb-related factors), we included interaction terms when considering two variables. Based on this approach, we constructed the candidate models shown in **Table 3**.

Model performance was evaluated using the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC). For each response variable, we selected the model with the lowest AIC and BIC values, as shown in **Table 4**. The selected models were referred to as the **Theft Model**, the **Crime Model**, and the **Rate Model**.

Table 3: Candidate Models

Model	Covariates	DF
car1	`num_spots`	4
car2	`price_pp`	4
car3	`num_spots * price_pp`	6
car4	`num_spots * price_pp + income_pc`	7
car5	`num_spots * price_pp + poverty`	7
car6	`num_spots * price_pp + poverty * income_pc`	9
sar1	`num_spots`	4
sar2	`price_pp`	4
sar3	`num_spots * price_pp`	6
sar4	`num_spots * price_pp + income_pc`	7
sar5	`num_spots * price_pp + poverty`	7
sar6	`num_spots * price_pp + poverty * income_pc`	9

Table 4: AIC and BIC Values for Models

number of theft			**number of crimes**			**ratio theft to crimes**		
Model	**AIC**	**BIC**	**Model**	**AIC**	**BIC**	**Model**	**AIC**	**BIC**
car1	1169.551	1178.488	car1	1380.417	1389.353	car1	-189.9745	-181.0381
car2	1187.944	1196.880	car2	1381.139	1390.075	car2	-180.9865	-172.0501
car3	1150.550	1163.955	car3	1380.934	1394.339	car3	-194.9684	-181.5637
car4	1151.818	1167.456	car4	1378.121	1393.760	car4	-229.5628	-213.9241
car5	1150.693	1166.332	car5	1375.425	1391.064	car5	-211.2674	-195.6286
car6	1145.254	1165.361	car6	1376.600	1396.707	car6	-233.9201	-213.8123
sar1	1167.001	1175.937	sar1	1387.016	1395.951	sar1	-213.8131	-186.2731
sar2	1184.837	1193.775	sar2	1378.456	1387.393	sar2	-188.8298	-179.8934
sar3	1149.936	1163.341	sar3	1377.700	1391.174	sar3	-197.4637	-184.0591
sar4	1151.252	1166.891	sar4	1376.476	1392.115	sar4	-229.5877	-213.9490
sar5	1150.250	1165.888	sar5	1390.030	1394.974	sar5	-211.1545	-195.5157
sar6	1144.421	1164.528	sar6	1374.867	1394.974	sar6	-233.9193	-213.8123

3.2 Resulting model

3.2.1 Model with Response Variable `num_theft`

According to the results of AIC and BIC, we select the model `sar6`. This model will be referred to as the **Theft Model**. The selected model for the number of thefts (`num_theft`) is specified as follows:

```
sar6 <- spautolm(num_theft ~ num_spots * price_pp + poverty * income_pc,
                 data = data,
                 family = "SAR",
                 listw = datab)
```

The interaction terms `num_spots * price_pp` and `poverty * income_pc` are included to capture joint effects of these variables.

Coefficient Estimation and Model Result The results of the selected SAR model (`sar6`) are as follows:

Table 5: SAR Model Coefficients and Summary Statistics

Variable	Estimate	Std. Error	z value	Pr($>\|z\|$)
(Intercept)	790.1029	724.6654	1.0903	0.2757
num_spots	-5.2323	3.2407	-1.6167	0.1064
price_pp	7.7023	3.8725	1.9901	0.0675
poverty	-44.7104	21.6812	-2.0627	0.0627
income_pc	-0.0449	0.0163	-2.7576	0.0297
num_spots:price_pp	0.0920	0.0464	1.9837	0.0008
poverty:income_pc	0.0261	0.0125	2.0857	0.0079

Statistic	Value
Lambda	0.1021
LR Test Value	6.8139
p-value for Lambda	0.0090
Numerical Hessian Std. Error	0.0329
Log Likelihood	-563.2106
ML Residual Variance	678630
Residual Standard Deviation	822.42
Number of Observations	69
Number of Parameters Estimated	9
AIC	1144.4

The results indicate that `price_pp`, `poverty`, and `income_pc` have statistically significant effects on the number of thefts. The interaction terms `num_spots:price_pp` and `poverty:income_pc` also contribute significantly to the model. The spatial dependence parameter (λ) is positive and significant, confirming the presence of spatial autocorrelation. This model was selected based on the lowest AIC and BIC values compared to other candidate models.

3.2.2 Model with Response Variable `num_crimes`

According to the results of AIC and BIC, we select the model `sar5`. This model will be referred to as the **Crime Model**. The selected model for the number of thefts (`num_crime`) is specified as follows:

```
sar5 <- spautolm(num_crimes ~ num_spots * price_pp + poverty,
                 data = data,
                 family = "SAR",
                 listw = datab)
```

Coefficient Estimation and Model Result The results of the selected SAR model (`sar5`) are as follows:

Table 6: SAR Model Coefficients and Summary Statistics

Variable	Estimate	Std. Error	z value	Pr($>\|z\|$)
(Intercept)	-1193.2	2300.3	-0.5187	0.603966
num_spots	3.0689	15.192	0.2020	0.839909
price_pp	38.864	20.269	1.9174	0.055181
poverty	7.7551	6.7662	2.5940	0.009488
num_spots:price_pp	0.0769	0.1204	0.6388	0.522954

Statistic	Value
Lambda	0.11329
LR Test Value	11.199
p-value for Lambda	0.00081824
Numerical Hessian Std. Error	0.028288
Log Likelihood	-680.1955
ML Residual Variance	19731000
Residual Standard Deviation	4442
Number of Observations	69
Number of Parameters Estimated	7
AIC	1374.4

The results indicate that `price_pp` and `poverty` have a statistically significant effect on the number of crimes. The spatial dependence parameter (λ) is positive and significant, confirming the presence of spatial autocorrelation. This model was selected based on the lowest AIC and BIC values compared to other candidate models. The results of the **Crime Model** indicate that while the interaction term `num_spots:price_pp` was not statistically significant, we retained it in the model due to the project's assumption of considering the joint effects of these variables. However, further analysis showed that removing this interaction term led to a better model fit, as evidenced by improved AIC and BIC values.

3.2.3 Model with Response Variable `rate`

According to the results of AIC and BIC, we select the model `sar6`. This model will be referred to as the **Rate Model**. The selected model for the number of thefts (`rate`) is specified as follows:

```
sar6 <- spautolm(rate ~ num_spots * price_pp + poverty * income_pc,
                 data = data,
                 family = "SAR",
                 listw = datab)
```

The interaction terms `num_spots * price_pp` and `poverty * income_pc` are included to capture joint effects of these variables on the ratio of thefts to crimes.

Coefficient Estimation and Model Result The results of the selected SAR model (`sar6`) are as follows:

Table 7: SAR Model Coefficients and Summary Statistics

Variable	**Estimate**	**Std. Error**	**z value**	**Pr(>|z|)**
(Intercept)	1.7385e-01	2.8690e-02	6.0177	1.769e-09
num_spots	1.8384e-04	1.3509e-04	0.1767	0.8617
price_pp	1.0511e-04	1.8564e-04	0.5665	0.5728
poverty	2.7314e-03	1.6943e-03	1.6125	0.1071
income_pc	-2.9214e-07	1.9149e-07	-1.5250	0.0555
num_spots:price_pp	7.8325e-07	1.5860e-06	0.6112	0.5279
poverty:income_pc	-9.4012e-08	2.5833e-08	-2.5833	0.0098

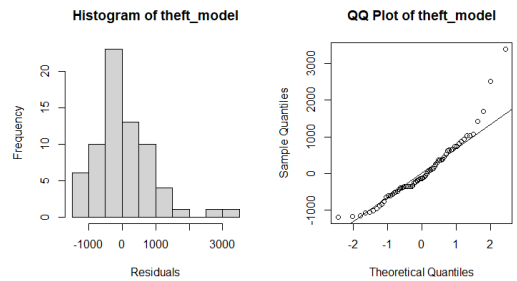
Statistic	**Value**
Lambda	0.005813
LR Test Value	0.021161
p-value for Lambda	0.88434
Numerical Hessian Std. Error	0.039026
Log Likelihood	125.9596
ML Residual Variance	0.00152
Residual Standard Deviation	0.03887
Number of Observations	69
Number of Parameters Estimated	9
AIC	-233.9193

The results indicate that the interaction term `poverty:income_pc` has a statistically significant effect on the ratio of thefts to crimes. The spatial dependence parameter (λ) is not statistically significant, suggesting that spatial autocorrelation is minimal in this model.

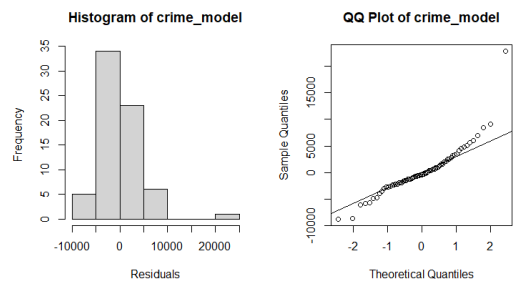
4 Discussion/Conclusion

4.1 Diagnostic

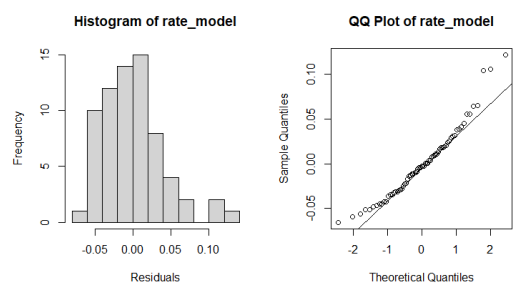
Normality of Residuals The residuals of all the fitted models were analyzed to check for normality and constant variance. Visual diagnostics were performed using histograms and Q-Q (quantile-quantile) plots for each model: the **Theft Model**, the **Crime Model**, and the **Rate Model**. The results are shown in Figure ??. These diagnostic plots suggest that the residuals for all three models meet the assumption of normality.



(a) Theft model's residuals



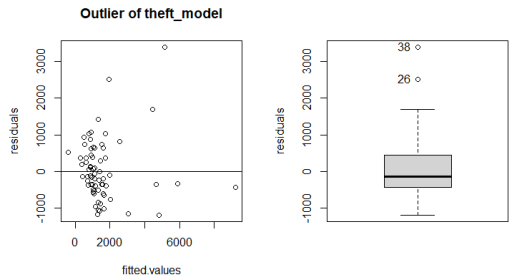
(b) Crime model's residuals



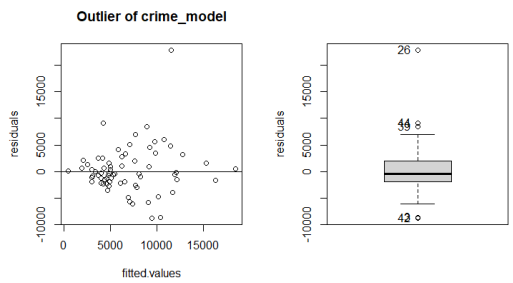
(c) Rate model's residuals

Figure 3: Normality of Residuals

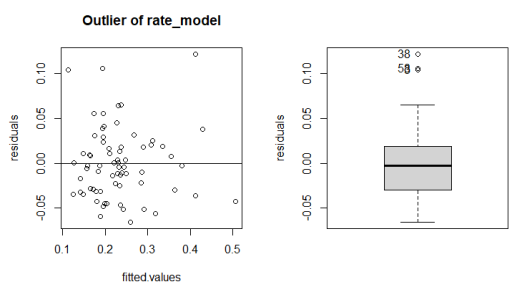
Outlier Detection During the diagnostic analysis of the fitted models, we identified several outliers in the residuals. The number of outliers detected for each model is as follows: 2 outliers detected in theft model, 5 outliers detected in crime model, and 3 outliers detected in rate model. These outliers were identified through residual analysis, including visual inspection of residual plots and statistical measures such as boxplots and standardized residuals.



(a) Theft model's residuals



(b) Crime model's residuals



(c) Rate model's residuals

Figure 4: Outlier Detection

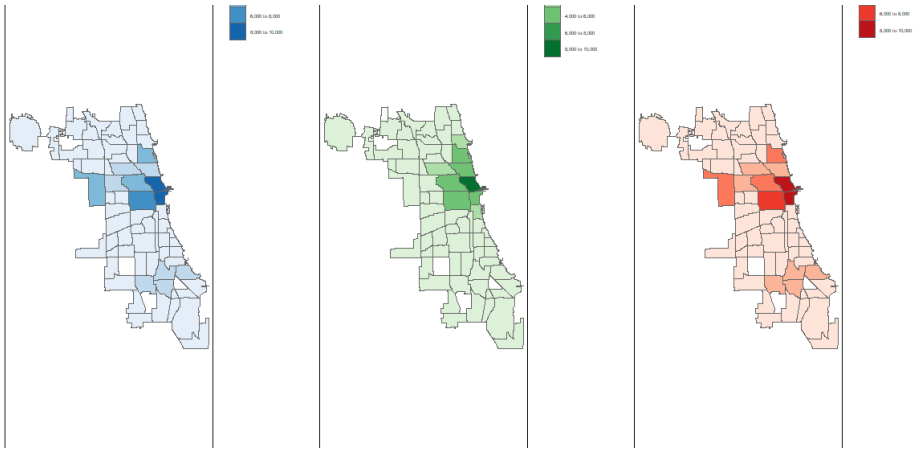
Moran's I-test of Residuals We conducted Moran's I tests on the residuals of the fitted models to assess whether spatial autocorrelation remained. The results for each model are summarized in Table 8. For all three models, the p-values are greater than 0.05, indicating that the spatial autocorrelation was effectively captured by the models.

Table 8: Moran's I Test Results for Model Residuals

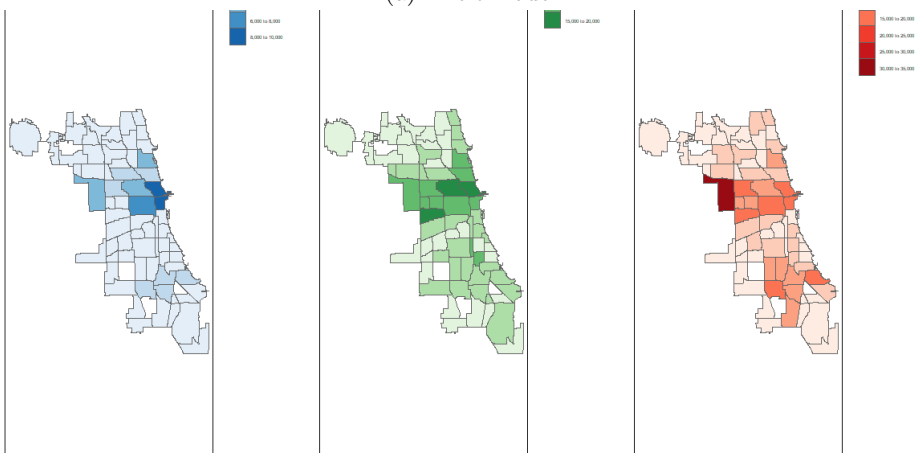
Model	Std. Deviate	p-value	Moran's I	Expectation	Variance
Theft Model	0.3819	0.3513	0.0147	-0.0147	0.0059
Crime Model	0.1541	0.4387	-0.0033	-0.0147	0.0050
Rate Model	-0.2755	0.6086	-0.0036	-0.0147	0.0061

The results show that all three models have p-values greater than 0.05, indicating that there is no significant spatial autocorrelation remaining in the residuals. This suggests that the models have successfully accounted for spatial dependence.

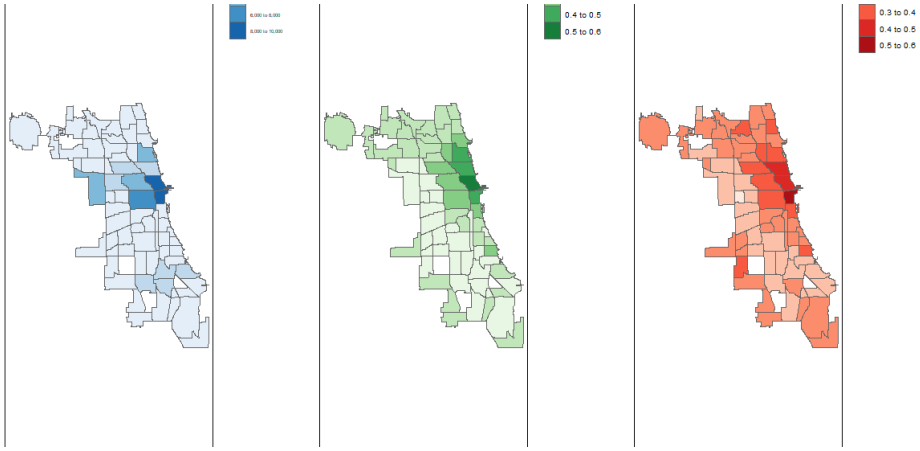
Visualization To better understand the performance of the models, we visualized the results using the `tmap` package by comparing the actual map, fitted map, and predicted map.



(a) Theft model



(b) Crime model



(c) Rate model

Figure 5: Blue: Exact map, Green: Fitted map, Red: Predicted map

In the **Theft Model** and **Rate Model**, the fitted and predicted maps closely resemble the actual map, capturing the spatial patterns effectively. However, in the **Crime Model**, while the overall trends are well-reflected, there are fitting difficulties in areas where the `rate` is high.

These visual comparisons highlight the effectiveness of the models and reveal areas where improvements may be necessary, particularly for regions with high crime rates.

4.2 Conclusion

In this study, we developed models to analyze theft numbers, crime numbers, and the ratio of thefts to crimes based on data observed in Chicago. The models were designed to incorporate factors related to Airbnb, including the number of Airbnb spots, the price per person for Airbnb rentals, the poverty rate, and income per capita. Our findings indicate that all three models perform well when some of variables are included, demonstrating their explanatory power.

Through diagnostic analysis, we found that the residuals of each model show the following characteristics: Although a few outliers were detected in the residuals, the residuals generally follow a normal distribution. Moran's I test on the residuals confirmed that the models effectively capture spatial autocorrelation, as no significant spatial patterns remained in the residuals.

To validate the performance of the models, we visualized the actual, fitted, and predicted maps. The fitted and predicted maps closely resemble the actual map, particularly for the theft and rate models. The crime model captures the overall trends well, although slight fitting difficulties were observed in regions with high theft-to-crime ratios.

These results suggest that our models provide a reliable representation of spatial patterns in theft and crime data in Chicago. The inclusion of Airbnb-related factors and poverty improves the models' ability to explain variations in crime rates. This study highlights the importance of considering spatial and socioeconomic factors in urban crime analysis.

5 Appendix

5.1 Code

```r
library(spdep)
library(nimble)
library(sf)
library(tidyverse)
library(nimble)
library(mapview)
library(spatialreg)
library(car)
library(tmap)
library(ggplot2)

setwd("C:/Users/kimsuna/Documents/Rdata/spatial/airbnb_Chicago/airbnb_Chicago")
#setwd("C:/Users/SUNA/Documents/R/spatstat/airbnb_Chicago")

#preprocessing
data<-sf::st_read("airbnb_Chicago_2015.shp")
data=data[!is.na(data$price_pp),]
data$rate=data$num_theft/data$num_crimes

#visualize the data
mapview(data, zcol="price_pp")
mapview(data, zcol="num_spots")
mapview(data,zcol="poverty")
mapview(data,zcol="income_pc")


mapview(data, zcol="num_theft")
mapview(data, zcol="num_crimes")
mapview(data, zcol="rate")



#moran test
data_nb <-spdep::poly2nb(data, queen=TRUE)
dataw <-nb2listw(data_nb, style= "W")
(moran <-moran.test(data$num_theft,dataw, alternative = "greater"))
(moranMC <-moran.mc(data$num_theft,dataw, nsim=999))
(moran <-moran.test(data$num_crimes,dataw, alternative = "greater"))
(moranMC <-moran.mc(data$num_crimes,dataw, nsim=999))
(moran <-moran.test(data$rate,dataw, alternative = "greater"))
```

```
(moranMC <-moran.mc(data$rate,dataw, nsim=999))
```

Listing 1: EDA

```
#theft modeling
datab <- spdep::nb2listw(data_nb, style = "B")
car1 <- spautolm(num_theft~num_spots, data=data,
                 family='CAR',listw=datab)
car2 <- spautolm(num_theft~price_pp, data=data,
                 family='CAR',listw=datab)
car3 <- spautolm(num_theft~num_spots*price_pp, data=data,
                 family='CAR',listw=datab)
car4 <- spautolm(num_theft~num_spots*price_pp+income_pc, data=data,
                 family='CAR',listw=datab)
car5 <- spautolm(num_theft~num_spots*price_pp+poverty, data=data,
                 family='CAR',listw=datab)
car6 <- spautolm(num_theft~num_spots*price_pp+poverty*income_pc, data=data,
                 family='CAR',listw=datab)

sar1 <- spautolm(num_theft~num_spots, data=data,
                 family='SAR',listw=datab)
sar2 <- spautolm(num_theft~price_pp, data=data,
                 family='SAR',listw=datab)
sar3 <- spautolm(num_theft~num_spots*price_pp, data=data,
                 family='SAR',listw=datab)
sar4 <- spautolm(num_theft~num_spots*price_pp+income_pc, data=data,
                 family='SAR',listw=datab)
sar5 <- spautolm(num_theft~num_spots*price_pp+poverty, data=data,
                 family='SAR',listw=datab)
sar6 <- spautolm(num_theft~num_spots*price_pp+poverty*income_pc, data=data,
                 family='SAR',listw=datab)
cbind(AIC(car1,car2,car3,car4,car5,car6,sar1,sar2,sar3,sar4,sar5,sar6),BIC(car1,car2,
    car3,car4,car5,car6,sar1,sar2,sar3,sar4,sar5,sar6))

theft_model=sar6
summary(theft_model)
```

Listing 2: Theft modeling

```
#crime modeling
car1 <- spautolm(num_crimes~num_spots, data=data,
                  family='CAR',listw=datab)
car2 <- spautolm(num_crimes~price_pp, data=data,
                 family='CAR',listw=datab)
car3 <- spautolm(num_crimes~num_spots*price_pp, data=data,
                 family='CAR',listw=datab)
car4 <- spautolm(num_crimes~num_spots*price_pp+income_pc, data=data,
                 family='CAR',listw=datab)
car5 <- spautolm(num_crimes~num_spots*price_pp+poverty, data=data,
                 family='CAR',listw=datab)
car6 <- spautolm(num_crimes~num_spots*price_pp+poverty*income_pc, data=data,
                 family='CAR',listw=datab)


sar1 <- spautolm(num_crimes~num_spots, data=data,
                 family='SAR',listw=datab)
sar2 <- spautolm(num_crimes~price_pp, data=data,
                 family='SAR',listw=datab)
sar3 <- spautolm(num_crimes~num_spots*price_pp, data=data,
                 family='SAR',listw=datab)

sar4 <- spautolm(num_crimes~num_spots*price_pp+income_pc, data=data,
                 family='SAR',listw=datab)

sar5 <- spautolm(num_crimes~num_spots*price_pp+poverty, data=data,
                 family='SAR',listw=datab) ##

sar6 <- spautolm(num_crimes~num_spots*price_pp+poverty*income_pc, data=data,
                 family='SAR',listw=datab)
cbind(AIC(car1,car2,car3,car4,car5,car6,sar1,sar2,sar3,sar4,sar5,sar6),BIC(car1,car2,
    car3,car4,car5,car6,sar1,sar2,sar3,sar4,sar5,sar6))
```

```r
crime_model=sar5
```

Listing 3: Crime modeling

```r
#rate modeling
car1 <- spautolm(rate~num_spots, data=data,
                 family='CAR',listw=datab)
car2 <- spautolm(rate~price_pp, data=data,
                 family='CAR',listw=datab)
car3 <- spautolm(rate~num_spots*price_pp, data=data,
                 family='CAR',listw=datab)
car4 <- spautolm(rate~num_spots*price_pp+income_pc, data=data,
                 family='CAR',listw=datab)
car5 <- spautolm(rate~num_spots*price_pp+poverty, data=data,
                 family='CAR',listw=datab)
car6 <- spautolm(rate~num_spots*price_pp+poverty*income_pc, data=data,
                 family='CAR',listw=datab)


sar1 <- spautolm(rate~num_spots, data=data,
                 family='SAR',listw=datab)
sar2 <- spautolm(rate~price_pp, data=data,
                 family='SAR',listw=datab)
sar3 <- spautolm(rate~num_spots*price_pp, data=data,
                 family='SAR',listw=datab)

sar4 <- spautolm(rate~num_spots*price_pp+income_pc, data=data,
                 family='SAR',listw=datab)
sar5 <- spautolm(rate~num_spots*price_pp+poverty, data=data,
                 family='SAR',listw=datab)
sar6 <- spautolm(rate~num_spots*price_pp+poverty*income_pc, data=data,
                 family='SAR',listw=datab)

cbind(AIC(car1,car2,car3,car4,car5,car6,sar1,sar2,sar3,sar4,sar5,sar6),BIC(car1,car2,
    car3,car4,car5,car6,sar1,sar2,sar3,sar4,sar5,sar6))
rate_model=sar6
```

Listing 4: Rate modeling

```r
#residual normality
model_list <- list(theft_model = theft_model,
                   crime_model = crime_model,
                   rate_model = rate_model)

par(mfrow = c(1, 2))
par(pty = "s")

for (model_name in names(model_list)) {
  model <- model_list[[model_name]]
  hist(model$fit$residuals,
       main = paste("Histogram of", model_name),
       xlab = "Residuals")
  qqnorm(model$fit$residuals,
         main = paste("QQ Plot of", model_name))
  qqline(model$fit$residuals)
}



#outlier
par(mfrow=c(1,2))

for (model_name in names(model_list)) {
  model <- model_list[[model_name]]
  plot(model$fit$fitted.values, model$fit$residuals,main=paste("Outlier of", model_
    name),xlab="fitted.values",ylab="residuals")
  abline(h=0)
  Boxplot(model$fit$residuals, id.method="y",ylab="residuals")
}

boxplot.stats(theft_model$fit$residuals)$out
```

```r
boxplot.stats(crime_model$fit$residuals)$out
boxplot.stats(rate_model$fit$residuals)$out



# Moran's I test
moran.test(theft_model$fit$residuals, dataw)
moran.test(crime_model$fit$residuals, dataw)
moran.test(rate_model$fit$residuals, dataw)
```

Listing 5: Diagnosis

```r
#Visualization
#theft model
fitted_values <- fitted(theft_model)
predicted_values <- fitted_values + residuals(theft_model)

data$fitted_values <- fitted_values
data$predicted_values <- predicted_values

tmap_mode("plot")

actual_map <- tm_shape(data) +
  tm_polygons("num_theft", title = "Actual Theft Rates", palette = "Blues") +
  tm_layout(title = "Actual Theft Rates", legend.outside = TRUE)

fitted_map <- tm_shape(data) +
  tm_polygons("fitted_values", title = "Fitted Theft Rates", palette = "Greens") +
  tm_layout(title = "Fitted Theft Rates", legend.outside = TRUE)

predicted_map <- tm_shape(data) +
  tm_polygons("predicted_values", title = "Predicted Theft Rates", palette = "Reds") +
  tm_layout(title = "Predicted Theft Rates", legend.outside = TRUE)

tmap_arrange(actual_map, fitted_map, predicted_map)


# crime model
fitted_values <- fitted(crime_model)
predicted_values <- fitted_values + residuals(crime_model)

data$fitted_values <- fitted_values
data$predicted_values <- predicted_values

tmap_mode("plot")

actual_map <- tm_shape(data) +
  tm_polygons("num_theft", title = "Actual Theft Rates", palette = "Blues") +
  tm_layout(title = "Actual Theft Rates", legend.outside = TRUE)

fitted_map <- tm_shape(data) +
  tm_polygons("fitted_values", title = "Fitted Theft Rates", palette = "Greens") +
  tm_layout(title = "Fitted Theft Rates", legend.outside = TRUE)

predicted_map <- tm_shape(data) +
  tm_polygons("predicted_values", title = "Predicted Theft Rates", palette = "Reds") +
  tm_layout(title = "Predicted Theft Rates", legend.outside = TRUE)

tmap_arrange(actual_map, fitted_map, predicted_map)


# rate model
fitted_values <- fitted(rate_model)
predicted_values <- fitted_values + residuals(rate_model)

data$fitted_values <- fitted_values
data$predicted_values <- predicted_values

tmap_mode("plot")

actual_map <- tm_shape(data) +
```

```
  tm_polygons("num_theft", title = "Actual Theft Rates", palette = "Blues") +
  tm_layout(title = "Actual Theft Rates", legend.outside = TRUE)

fitted_map <- tm_shape(data) +
  tm_polygons("fitted_values", title = "Fitted Theft Rates", palette = "Greens") +
  tm_layout(title = "Fitted Theft Rates", legend.outside = TRUE)

predicted_map <- tm_shape(data) +
  tm_polygons("predicted_values", title = "Predicted Theft Rates", palette = "Reds") +
  tm_layout(title = "Predicted Theft Rates", legend.outside = TRUE)

tmap_arrange(actual_map, fitted_map, predicted_map)
```

Listing 6: Visualization

References

[1] Charles C. Lanfear and David S. Kirk. *The promise and perils of the sharing economy: The impact of Airbnb lettings on crime*. Criminology, 2024, 62 (4), pp. 1023-1046. https://doi.org/10.1111/1745-9125.12383

[2] Stephen P. Kaluzny, Silvia C. Vega, Tamre P. Cardoso, and Alice A. Shelly. *S+SpatialStats: User's Manual for Windows® and UNIX®*. Springer, 1998. https://link.springer.com/book/10.1007/978-1-4615-7826-0

[3] Alain F. Zuur, Elena N. Ieno, and Graham M. Smith. *Analyzing Ecological Data*. Springer, 2007. https://link.springer.com/book/10.1007/978-0-387-45972-1